

Experimental apparatus for universal spin-motion control in multi-ion registers



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1

Towards Hybrid System Control and Quantum Error Mitigation

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Having established and characterised the experimental apparatus and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (§ ??), the single addressing system (§ ??), and the developed control over both the discrete qubits, as well as the continuous motional modes.

In § 1.1 we establish a set of unitary operations required for universal continuous-variable (CV) control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. Following on from recent progress in spin-conditioned

bosonic operations demonstrated within our group [1–5], we are interested in developing the required addressing capabilities to scale to multiple qubits and motional modes. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will allow substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [6–8] — the specific method considered here. We provide an initial demonstrations of this protocol applied to a discrete-variable only system, and explore the performance under both local depolarising and correlated noise due to a common fluctuating control field (see § ?? for details).

We conclude this chapter with a discussion on the application of virtual distillation to hybrid systems, presenting a scalable method within reach of our apparatus.

1.1 Hybrid Spin-Oscillator Control in Trapped Ions

CV quantum computation [9] offers an alternative strategy for encoding quantum information, utilising quantum harmonic oscillators, termed *qumodes*, as the fundamental quantum object, rather than discrete variable (DV) systems, e.g. the two-level qubit. A single ideal qumode possesses an infinite dimensional Hilbert space, allowing for high (in this case arbitrarily high) density encoding of information in physical information carriers. This opens an alternative avenue for scaling QIPs, requiring dramatically less physical carriers compared to qubit-only systems. The inclusion of both CV and DV subsystems, termed the hybrid model [10], retains the convenient control and readout of discrete qubits while providing the larger state

space and native interactions available to qumodes. For example, reference [11] considers a factoring algorithm with the required physical resources independent of the number to be factored. Remarkably, this scheme requires only a single qubit and three oscillators, however requires prohibitively large energy states of the oscillator. Experimentally, we will not be able to realise arbitrarily high energy oscillator states, but it is still an open question of what information density is possible in accessible engineered quantum oscillator.

Hybrid spin-oscillator computation has been studied in the context of superconducting cavity-QED [12–16], neutral atom in optical tweezers [17–19], and in the trapped-ion platform [20–22]. We shall focus on the trapped-ion implementation, where the DV system is the collection of qubits embedded in the electron energy levels, and the qumodes are the shared secular motional modes of the ion crystal within the trapping pseudopotential.

Trapped ions benefit from multiple native methods to drive, and couple qubit and qumode: The qumodes may be directly driven through parametric trap potential modulation [23–25], or through near mode-resonance RF fields; the qubit may be driven either by optical fields (in the case of optical qubit encoding), or through microwaves (in the case of ground state encoding) resonant with the energy-state transition; the qubit and qumode may be coupled through qubit-state dependent potential shifts [26, 27], or through optical fields tuned near to motional-mode frequency sidebands of the qubit-carrier transition, for example through the Jaynes-Cummings interaction (§ ??), or the Mølmer-Sørensen (MS) SDF [28]. Taken together, these native control and coupling mechanisms, combined with the long coherence times of both the qubit and qumode, make the trapped-ion platform well-suited for exploring hybrid CV–DV quantum computation.

The apparatus developed in this thesis has demonstrated good qubit and radial motional coherence times, provides access to multiple qubits and qumodes, and includes a single-addressing system for selectively driving their coupling. These capabilities make it well-placed for studying methods for scaling hybrid-system control.

Interaction name	Oscillator H	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω
Displacement	$ae^{i\phi} + a^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_x$	Ω_α
Parity	$aa^\dagger e^{i\phi} + a^\dagger a e^{-i\phi}$	✓	1	+1	$\hat{\sigma}_z$	Ω_2
Beam splitter	$ab^\dagger e^{i\phi} + a^\dagger b e^{-i\phi}$	✓	2	+1	$\hat{\sigma}_z$	Ω_2
Squeezing	$a^2 e^{i\phi} + a^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_z$	Ω_2
Trisqueezing	$a^3 e^{i\phi} + a^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_x$	Ω_3

Table 1.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. Using the protocol presented in reference [20], here termed the “Two-SDF” method, we show the required relative detuning setting, m , and the resulting spin conditioning, $\hat{\sigma}$, and coupling strength Ω of the resulting interaction. Ω_α is the typical SDF strength given by equation ???. The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = 2\Omega_\alpha\Omega_{\alpha'}/\Delta$, $\Omega_3 = 2\Omega_\alpha^2\Omega_{\alpha'}/\Delta^2$.

1.1.1 Theory

Gateset for a Hybrid System

For any computing paradigm, we must establish a set of interactions required for universality. In DV-only systems, quantum universality may be realised by a finite set of Clifford operations plus one non-Clifford gate [29, 30]. CV-only universality can analogously be realised by a finite set of Gaussian operations plus one non-Gaussian gate [9, 31]. The term Gaussian is understood in the position-momentum representation of the qumode, where the bare oscillator Hamiltonian is given by,

$$H_{\text{osc},0} = \hbar\omega_{\text{osc}}(\hat{x}^2 + \hat{p}^2), \quad (1.1)$$

with quadratures,

$$\hat{x} = \frac{1}{2}(a + a^\dagger), \quad (1.2)$$

$$\hat{p} = \frac{-i}{2}(a - a^\dagger), \quad (1.3)$$

where $a^\dagger$, a are the creation and annihilation ladder operators of the mode with frequency, ω_{osc} . Gaussian operators are then the set of interactions containing at most quadratic terms in $\hat{x}$ and $\hat{p}$, therefore mapping Gaussian input states to Gaussian output states.

The condition for universality of a hybrid CV-DV is somewhat ambiguous as both

the CV and DV subsystems are themselves sufficient for universality, however this presents an opportunity: the freedom in selecting the chosen gate-set for universality, combining both Clifford, non-Clifford, Gaussian, non-Gaussian, and spin-oscillator interactions together, allows for selecting efficient, native, and simply accessed interactions of the physical platform. We consider the over-complete instruction set presented in Table 1.1, consisting of spin-conditioned Gaussian and non-Gaussian interactions. The interactions on the bosonic subsystems are themselves universal, but the spin-conditioning provides a bridge between CV and DV subsystems, allowing for readout and control to be implemented on ancilla qubits. Specifically, we consider a cubic CV instruction set [9, 12, 32], taking the Gaussian operations and supplementing with a non-Gaussian “tri-squeezing” gate, with Hamiltonian,

$$H_T \propto a^3 + a^{\dagger 3}. \quad (1.4)$$

Spin-conditioned tri-squeezing has been shown previously in our group [1], utilising the protocol given in reference [20], here termed the “two-SDF” method.

The remaining non-linear Gaussian operations; squeezing, parity, and beam-splitter, may be generated through the same “two-SDF” mechanism, which we shall detail in the following section, and have also been previously demonstrated [1, 3, 4, 33]. The displacement operation can be realised either spin-independently by RF fields near resonance to the motional-mode frequencies (§ ??), or spin-conditioned using various SDF implementations, such as by the MS method using bichromatic fields (§ ??), or by static Stark-shift gradients due to an optical tweezer [27].

As highlighted in this section, the required unitaries for universal hybrid CV-DV control may all be accessed by a common protocol: The “two-SDF” method. This greatly simplifies experimental hardware and control complexity. We shall give an overview of this method in the following section, and provide a practical route and demonstration utilising four optical frequencies with fixed relative detunings and phases.

- Do we need to define each interaction, sqz, parity, bs?
- Could give flow fields for single-mode operators (could be a nice figure, and I like that representation).

Gate Implementation

To implement the unitaries given in table 1.1, and to retain spin-dependent control of these interactions to fully utilising the large hybrid state space, we use the “two-SDF” method proposed in reference [20]. The non-linear bosonic interactions are effectively driven through applying two linear spin-dependent forces with non-commuting spin conditioning, simultaneously on a single qubit. The two simultaneous SDFs, with Hamiltonians and parameters labelled by subscripts α , and β , are applied off-resonantly relative to motional modes with creation operators $\hat{a}_\alpha^\dagger$ and $\hat{a}_\beta^\dagger$, respectively. The resulting Hamiltonian is given generally by,

$$\hat{H}_{\text{2SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (1.5)$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (1.6)$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right), \quad (1.7)$$

where the individual SDF strengths are Ω_i , the detuning from each mode is Δ_i , the motional phase is ϕ_i , and we ensure the spin-conditioning anticommute $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 0$. The unitary dynamics of $\hat{H}_{\text{2SDF}}$ may be evaluated by the Magnus expansion [34], with details given in references [3, 4, 20], and in appendix A.8. To lend some intuition on the resulting physics, the Magnus expansion expresses the generator of the unitary evolution as an infinite series of terms, with successive terms proportional to increasingly higher-order nested commutators of the Hamiltonian evaluated at different times. Due to the non-commuting spin conditioning, this series does not truncate, leading to arbitrarily high non-linear bosonic interactions effectively driven via the application of only two linear SDFs.

To utilise specific non-linear terms within this infinite series, matching the desired interactions given in table 1.1, we constrain the relative detunings of the two SDFs

to $\Delta_\beta = m\Delta_\alpha$, where m is some integer given in the table. This immediately brings components of the Magnus terms matching the desired non-linear $\hat{H}_{\text{osc}}$ into resonance. Undesired higher-order terms also brought into resonance may be arbitrarily suppressed as the effective coupling of the n th Magnus term is $\Omega_{\text{eff}} \propto (\Omega_\alpha/\Delta)^{n+1}$ (assuming $\Omega_\alpha = \Omega_\beta$). The remaining error terms are far off-resonant, and may also be further suppressed by ensuring $\Delta \gg \Omega_\alpha\Omega_\beta$, and utilising appropriate smooth amplitude and phase envelopes for the pulses.

In this regime, the unitary dynamics of $\hat{H}_{\text{2SDF}}$ are well approximated by the evolution of time-independent effective Hamiltonians, which for Gaussian operations ($|m| = 1$) are of the form,

$$\hat{H}_{\text{eff,G}} = \frac{\hbar\Omega_\alpha\Omega_\beta}{2\Delta}\epsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma(ab^{(\dagger)}ie^{i\phi} + h.c.), \quad (1.8)$$

where $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbols. If $a_\alpha^\dagger = a_\beta^\dagger$, the SDFs are detuned from the same motional mode, leading to squeezing, or parity effective interactions, while acting on two-modes leads to the beam splitter interaction. For the tri-squeezing interaction ($m = -2$) the effective Hamiltonian is,

$$\hat{H}_{\text{eff,T}} = \frac{\hbar\Omega_\alpha^2\Omega_\beta}{4\Delta^2}\hat{\sigma}_\beta(a^3e^{i\phi} + a^{\dagger 3}e^{-i\phi}), \quad (1.9)$$

where we have set $a_\alpha^\dagger = a_\beta^\dagger$, as we act on a single mode.

Frequency and spin-conditioning control of two SDFs therefore allow for the generation of the entire proposed hybrid CV-DV gate-set. After a discussion on how we can characterise the oscillator state, we shall present the experimental scheme for generating these interactions, and demonstrate the creation of a squeezed state by this method.

Readout

With the above methods we may generate interesting non-classical bosonic states, now we provide a method for characterising them.

In contrast to a two-level system, the state of an oscillator may not be directly readout through state dependent fluorescence. This is due to the much larger

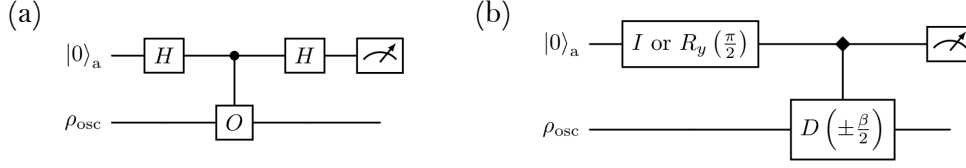


Figure 1.1: State tomography for a single mode through measurement of an ancilla qubit. (a) The Hadamard test method for measuring the expectation value of an arbitrary operator, $\hat{O}$, applied to the oscillator state, utilising an ancilliary qubit. (b) A Hadamard test circuit for measuring the characteristic function, $\chi(\beta)$, of the oscillator using native single qubit rotations and σ_x -conditioned spin-dependent forces (symmetric control represented by the diamond symbol) [35].

state-space of the qumode, and the technical difficulty in direct motional readout on the ion-trap platform. To solve these issues, a modular readout scheme is used, involving indirect readout of the oscillator state via a discrete collection of controlled unitaries subject to an ancillary qubit, and computational basis measurements of that ancilla.

The information is mapped to the ancilla by a Hadamard test [36], shown in figure 1.1 (a). Here the expectation of value due to the operator $\hat{O}$ on the oscillator, $\langle \hat{O} \rangle = \text{Tr}(\hat{O}\rho_{osc})$, is recovered through exclusive measurements of the ancilla qubit after qumode-dependent phase “kick-backs” are applied. The Hadamard test has the convenient relation between computational basis measurements of the ancilla, and the applied operator, $\langle \hat{\sigma}_z \rangle = \text{Re}[\text{Tr}(\hat{O}\rho_{osc})]$.

Full state tomography may now be formulated by repeat measurements of the conditional displacement operation, $\hat{O} = e^{-i\theta}\hat{D}(\beta)$, to recover the characteristic function, defined as $\chi = \text{Tr}(D(\beta)\rho_{osc})$, β is a complex parameter with displacement magnitude, $|\beta|$, and displacement axis, $\arg(\beta)$. Final computational state measurements of the Hadamard test circuit then yield,

$$\langle \sigma_z \rangle = \text{Re}[e^{-i\theta}\text{Tr}(D(\beta)\rho_{osc})], \quad (1.10)$$

$$= \cos(\theta)\text{Re}[\chi] + \sin(\theta)\text{Im}[\chi], \quad (1.11)$$

where θ may be set to 0 or $\pi/2$ to recover the real or imaginary part of the characteristic function respectively. Each measurement provides only 1-bit of

information on the (continuous) oscillator state, thus to approximate a useful reconstruction, many samples with a range of β values must be collected.

Once the characteristic function has been measure, it can be converted to other familiar representations of the oscillator, such as in the Fock-state basis, or Wigner representation [37–39]. In this work, we shall keep to the conveniently measured characteristic function.

To give two concrete example, the vacuum state characteristic function is the circularly symmetric Gaussian state,

$$\chi_0 = \exp\left(\frac{-|\beta|^2}{2}\right), \quad (1.12)$$

while the squeezed state, with squeezing parameter $\xi = re^{i\phi}$, breaks this symmetry due to a contraction on one axis, and a corresponding expansion on the other [40],

$$\chi_s = \exp\left(\frac{-|\beta|^2}{2} \left(e^{2r} \cos^2(\phi') + e^{-2r} \sin^2(\phi')\right)\right), \quad (1.13)$$

where ϕ' is the angle between squeezing axis and probe axis, $\phi' = \arg(\beta) - \frac{\phi}{2}$.

Experimentally the $\hat{\sigma}_x$ conditioned spin-dependent force are simply accessed by the bichromatic field method discussed in § ?? . The Hadamard test circuit is decomposed to native gates, shown in figure 1.1 (b), first described in reference [35]. The proceeding I or $R_y(\pi/2)$ set the quadrature to measure; $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. This circuit provides a practical method for characterising a single qumode, which we shall use in the following experimental demonstration. We note, similar to the constraints of state characterisation discussed in § ?? , full tomographic reconstruction is intractable for multiple-qumodes due to the large state-space, even with reasonably truncated oscillator representations (e.g. imposing a Fock-state cut-off). In the case of characterising systems consisting of multiple qubits and qumodes, either reduced tomography [41], or simplified metrics of performance [42] must be used.

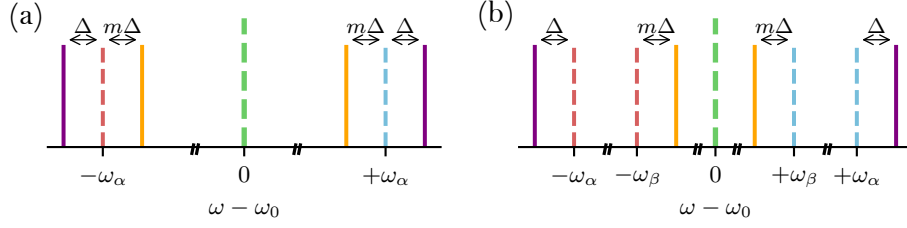


Figure 1.2: The quadchromatic field required for generating non-linear motional interactions via the two-SDF method. (a) For single-mode interactions, both SDF fields are detuned around the sidebands of the same motional mode, ω_α . Integer m is varied to select the desired interaction. (b) For two-mode interactions, each SDF is detuned from a different motional mode.

1.1.2 Experimental Demonstration

In our group, we have previously demonstrated higher order squeezing [1], superpositions of non-linear bosonic states [2], beam-splitter [33], and two-mode squeezing interactions. However, due to the constraints of the previous experimental apparatus we were limited to global addressing fields, preventing the application of these hybrid interactions to more than two qubits. Here we demonstrate the creation and characterisation of a squeezed state on the newly constructed apparatus, described in chapter ??, utilising the single-qubit addressing system. Single addressing both increases the power efficiency of the interactions, and allows the control to select which qubit is used to couple to the motion — this provides a scalable route to generating hybrid interactions on larger qubit, and qumode registers.

Two Spin Dependent Forces

The individual SDFs are generated by the same bichromatic field method described in § ?? — i.e. simultaneous off-resonant blue- (BSB) and red-sidebands (RSB) of motional mode ω_α are applied. As such, we require four optical fields with, frequency, phase, and amplitude control. The four required frequencies of the “quadchromatic” field are shown schematically in figure 1.2 with respect to the carrier frequency, ω_0 . Here, purple lines correspond to SDF_α , and orange lines to SDF_β . In this example we show the SDFs acting on a shared motional mode,

but in general they may act on different modes. Similar to the generation of the bichromatic field, we may produce this quadchromatic field through the application of four RF tones onto a single-pass (SP) AOM. These tones are generated using an arbitrary waveform generator (AWG)¹. We cannot control the absolute optical phase at the ion, however we can set relative phases between the four fields by applying phase offsets between the RF tones with respect to some fixed time point. As discussed in § ??, the spin phase of the SDFs is set by the absolute phase of the RSB and BSB, $\hat{\sigma}_\phi = \cos(\phi_s)\hat{\sigma}_x + \sin(\phi_s)\hat{\sigma}_y$. To ensure we saturate the non-commutativity of the spin conditioning of the two SDFs, we require $\pi/2$ phase difference between the two spin phases,

$$\phi_{\alpha,s} - \phi_{\beta,s} = \frac{\phi_{\alpha,\text{RSB}} + \phi_{\alpha,\text{BSB}}}{2} - \frac{\phi_{\beta,\text{RSB}} + \phi_{\beta,\text{BSB}}}{2} = \frac{\pi}{2}. \quad (1.14)$$

In the following demonstration we set $\phi_{\alpha,\text{RSB}} = \phi_{\alpha,\text{BSB}} = \pi/2$, $\phi_{\beta,\text{RSB}} = \phi_{\beta,\text{BSB}} = 0$, however we note that for composing multiple hybrid operations into a circuit, the resulting motional phase of the SDFs can lead to further constraints on the chosen RSB and BSB phases.

The intensity of the quadchromatic field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where t_{ramp} is the ramp duration. This ramp suppresses excess off-resonant excitations of the A_1 phase loop term given in the previous section, and so we must select its duration based on the characteristic scale, $\frac{1}{\Delta}$.

We aim to balance the power in each of the 4 tones. This may be calibrated using an optical pick-off before addressing the ion, directed onto a fast photodiode, and measuring the beatnote contrast when applying any pair of tones. In the following squeezing demonstration, we balance the tones comprising the individual SDFs, but do not explicitly balance the power between SDFs, as the desired interaction scales as $\Omega_\alpha\Omega_\beta$, any imbalance leads to non-optimal interaction efficiency, but unlike imbalance of the RSB and BSB, does not increase coherent error terms. We observe that the duty cycle of the SP AOM can dramatically shift the power imbalance

¹*Sinara Phaser* board

between all four tones, and so we take care ensuring that the total RMS power to drive one-, two-, and four-tones is the same as for the idle (no optical pulse) setting of the SP AOM, thus the only transient power drop is that of the ramp sequence.

Squeezing

- **potential plot: check ramp is sufficient by looking at σ_z dependence. If want ramp have rid 120238, 119279**

We demonstrate the effective squeezing interaction on a single ion and radial motional mode. We are limited to using the radial modes due to the geometry single addressing system with respect to the trap. We use the upper radial mode with frequency, $\omega_{upper} = 2.87$ MHz, and motional coherence $\tau_{upper} = 19(1)$ **rid 135596** given in § ???. The initial near-vacuum motional state is prepared by Doppler and sideband cooling, with a typical thermal fock state distribution with mean $\bar{n} = 0.03(1)$, as described in § ??. The spin state is initialised into the ground state by optical pumping, described in § ??. By setting the relative detuning between the SDFs to $m = -1$, and ensuring anti-commuting spin conditioning by setting the relative phases of the quadchromatic field according to equation 1.14, we realise a $\hat{\sigma}_z$ -conditioned squeezing interaction. As the spin-state is prepared in an eigenstate of the spin-conditioned squeezing, we expect squeezing on the qumode, without any change to the qubit state. We use a total quadchromatic beam power of 53 μ W, and set the base detuning, $\Delta = 2\pi$ 50 kHz. We suppress excess excitation of the off-resonant error terms by using a ramp of $t_{ramp} = 40$ μ s.

The state is characterised by the tomography method given in § 1.1.1, with a discretization set to collect a 35×35 grid of displacement values, β , and $N = 300$ repeat samples to reduce shot noise. The full dataset for one of the tomographic scans takes around 1 hour to collect, and so we interleave mode calibration experiments (§ ??) to run every 5 minutes.

The on-resonance splitting SDF used for tomography must have its strength, Ω_{SDF} ,

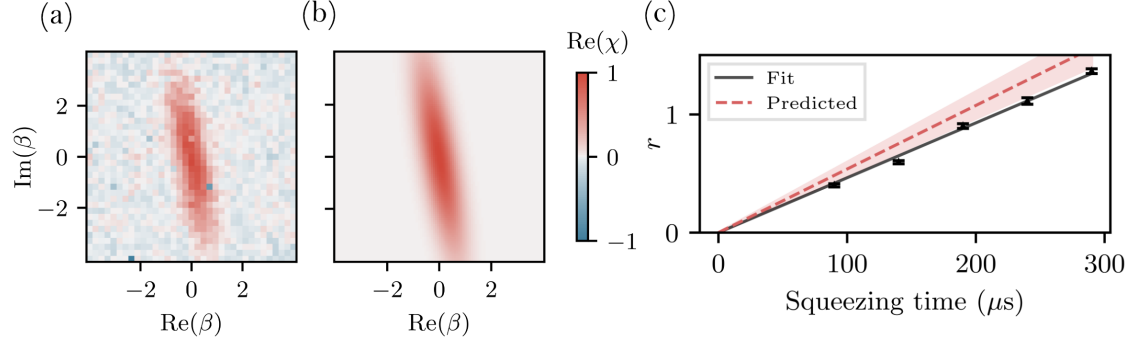


Figure 1.3: Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 1.1. (b) The simulated state taking fit parameters inferred by equation 1.13. (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

calibrated to infer the magnitude of β applied. We use a total bichromatic beam power of 40 μW , and through fitting the $\hat{\sigma}_z$ expectation for the time-dependent splitting of a vacuum state,

$$\langle \hat{\sigma}_z \rangle = e^{-2|\beta|^2}, \quad (1.15)$$

with

$$|\beta| = \frac{\Omega_{\text{split}} t}{2}. \quad (1.16)$$

We find $\Omega_{\text{split}} = 2\pi \cdot 7.2(4)$ kHz.

The pulse sequence for squeeze state generation and characterisation is given in figure ??, and the resulting real part of the characteristic function is shown in figure 1.3 (a). Fitting the data to equation 1.13 we may extract both the squeezing axis, ϕ , and the strength, r , where r is given by,

$$r = |\Omega_{\text{eff},S}|t = \frac{\Omega_\alpha \Omega_\beta}{2} t. \quad (1.17)$$

An example fit is shown in panel (b), with extracted $r = 0.90(2)$, for a quadchromatic FWHM pulse duration of $t_{\text{sqz}} = 190 \mu\text{s}$. Panel (c) displays the squeezing strength as a function of t_{sqz} , with fit values given by the points, and the black line is a linear fit. Assuming that the two SDF strengths are well balanced, $\Omega_\alpha = \Omega_\beta$, and taking the 53 μW used for the quadchromatic field compared to the SDF strength calibration

at 40 μW , we find an expected squeezing strength of $\Omega_{\text{eff},S} = 2\pi \cdot 0.86(9)$ kHz. This is shown as the red dashed line and shaded region. The fitted strength of $\Omega_{\text{eff},S} = 2\pi \cdot 0.74(1)$ kHz is lower than the expected, suggesting that either we are not sufficiently balancing the strengths of the two simultaneous SDFs, or that the power stabilisation routine for 4-tones is inaccurate. Further exploration is needed, including careful quantification of the individual SDF strengths constituting the quadchromatic field. However, this preliminary result confirms that the motional coherence, and stability of the single-addressing system is sufficient to create, and characterise non-classical motional states using the apparatus developed for this thesis.

Comment on heating rate

1.1.3 Outlook

Hybrid systems offer access to large Hilbert spaces, even with modest ion numbers. In this section we outlined a convenient set of native interactions for universal hybrid qubit-qumode gates. We experimentally demonstrate the squeezing interaction generated via the “two-SDF” method of reference [20]. This demonstration replicated previous work using the “two-SDF” method [1], however now shown on the newly built apparatus utilising a single-ion addressing system, and applied to the radial motional modes.

Enabled by the single-addressing, and improved motional coherence times of the new system, we are particularly interested in exploring short hybrid circuits composed of multiple interactions acting on an array of qubits and qumodes. Specifically, we are interested in utilising the hybrid system with increased robustness against errors through the use of quantum error mitigation methods applicable to both spin and oscillator subsystems. We outline one such method in the next section, with a preliminary demonstration on the spin subsystem.

Limitations

Using a single multi-ion chain to access multiple qubits and qumodes for scaling hybrid QC on the trapped-ion platform will be ultimately limited by spectral crowding of the motional modes, i.e. the average frequency spacing between neighbouring modes decreases as the number of accessible modes increases.

The “two-SDF” method requires frequency selectivity to drive bosonic interactions exclusively on the chosen mode. To selectively drive a single mode, the magnitude of the detuning Δ must follow $\Delta \ll \delta\omega$, where $\delta\omega$ is the frequency difference to the nearest neighbouring mode. As discussed in the “two-SDF” method section, the strength of each SDF must be less than the base detuning, $\Omega_\alpha \ll \Delta$, to well approximate the effective non-linear unitary dynamics. As a result of these two requirements, the mode-frequency splitting directly limits the attainable strength of the non-linear interaction in the adiabatic regime, resulting in increasing gate duration with increasing register size.

A possible mitigation would be to rather scale the hybrid system using a collection of fixed-length chains with adequate mode-frequency splitting, and to transport quantum information between these chains either through ion transport whilst maintaining mode and spin coherence [43], generating remote entanglement through photonic interconnects [44], or through inducing collective motional interactions between separate ion-chains [45].

Another mitigation strategy could come from the pulse-shaping approach demonstrated for use in fast two-qubit entangling gates in the non-adiabatic regime [46], i.e. when $\Delta > \delta\omega$. Introducing extra degrees of freedom in the pulse amplitude envelope could allow for driving the desired interaction on the selected modes, while returning all spectator modes to their initial state by the end of the pulse. However, we note that the accumulation of non-global geometric phases, and possible non-commutativity of interactions on shared qubit-oscillator states could prohibit the use of such a pulse-shaping scheme. Ultimately, the desired interaction, mode frequency spectra, and target gate duration will determine the dominant coherent

errors. We leave such an analysis for future work.

1.2 Virtual Distillation

With access to noisy, intermediate-scale quantum information processors, the natural question is whether useful computations may be executed, without requiring full fault-tolerant error correction codes. In this section, we explore a resource-efficient error mitigation technique, known as virtual distillation, that can dramatically suppress the effect of errors on the final measured outcome for expectation value problems.

This section starts with outlining the goals of quantum error mitigation (QEM), with the following section giving details on the virtual distillation method. We then present an experimental demonstration suppressing errors a single qubit system. We finish this section with a discussion on future plans, and the extension of virtual distillation to hybrid CV-DV systems.

1.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction (QEC) is to detect and correct these errors through embedding information in larger Hilbert spaces [qec_review]. Full QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore not feasible on current quantum hardware [<empty citation>] [is this fair to say?].

Furthermore, QEC is formulated for digital “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits, or analog quantum simulators consisting of tailored, often continuous, unitary evolution [nisq_review]. Particularly, phrasing the problem in terms of expectation values of an observable on a physical final state, rather than a sequence of measurements on a set of logical states, allows for a more efficient use of resources. As a tactile example, consider the problem of finding the ground state energy of a Hamiltonian. This can be done by preparing a trial state, and directly measuring the expectation value of the Hamiltonian on that state, building up statistics over many samples [variational_qc_review].

However, the impact of errors on uncorrected computation must be carefully managed. Uncertainty of the approximated expectation value due to averaging over a finite number of discrete samples, known as shot-noise, may be arbitrarily reduced by increasing the samples, N . However, due to imperfect execution of a given circuit, and coupling to the environment, there will be some systematic shift of the estimated average value ideal value in the noise-less limit. This shift, or bias, of the approximated observable may be reduced by the application of quantum error mitigation (QEM) techniques.

Reference [47] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the main distinction to QEC protocols: the physical noisy state is never corrected (often it is in fact degraded through the use of QEM) — rather the output from the noisy circuits is manipulated and postprocessed in such a way that we *effectively* recover an expectation value with suppressed deviation from the ideal value.

rethink narrative here

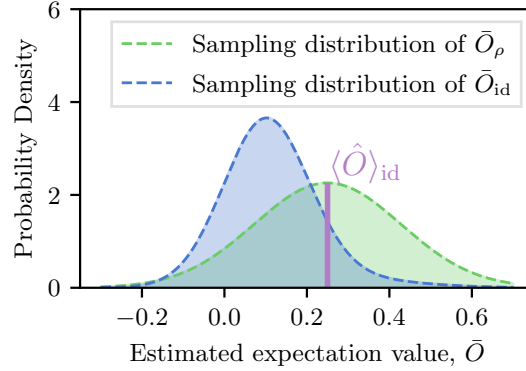


Figure 1.4: **Integrate figure into main text.** $\rho = 0.1\rho_{\text{id}} + 0.9\rho_{\text{err}}$. Blue curve is PDF for N shots, green is PDF for ideal shots, with $N_{\text{id}} = 0.1N$.

A schematic explanation of the goal of quantum error mitigation. Deviations of the measured expectation value of an operator $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by the ideal QEM method introduced in § 1.2.1. The underlying noisy state is given by $\rho = (1 - p)\rho_{\text{id}} + p\rho_{\text{err}}$. N shots are taken measuring $\text{Tr}(\hat{O}\rho)$. The ideal QEM flags and discards shots originating from ρ_{err} , leaving $(1 - p)N$ shots of the ideal state to estimate $\langle \hat{O} \rangle$.

- Finish argument about ideal QEM
- Summary of QEM method types (quick overview) and transfer sentence to VD purity type approach.
- Why VD:
 - operates on noisy physical states rather than requiring logical encoding;
 - is applicable to general Hilbert spaces, i.e. hybrid good
 - exchanges additional copies/sampling/circuit resources for reduced bias with fixed length (given copy and register size) derangement and measure at end
 - is therefore interesting for the multi-register and eventually hybrid architecture considered here.

Error mitigation protocols are diverse in their resource efficiency, assumptions on error-types, and scaling with system size. In this thesis, we consider the virtual distillation protocol [6–8], a QEM method, applicable to both general

Hilbert spaces, including both DV and CV systems, with few assumptions on the underlying noise model [48, 49]. The following section gives details on the mechanism and practical circuits.

1.2.2 Virtual Distillation Protocol

Virtual distillation is a purification-based QEM, utilising k -copies of a circuit to reduce the final bias of the measured expectation value of a noisy state, ρ . The term *purification* is used to signify that the method effectively approximates the pure state closest to ρ , with the underlying assumption that the dominant eigenvector of ρ approximates the ideal pure state, $\rho_{\text{id}} = |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}|$. The validity of this approximation is explored in detail in reference [48], where they derive a bound for the deviation between the ideal state and dominant eigenvector, termed the “coherent mismatch”, c . This error provides the noise-floor for purification based techniques. Favourably, for many error models and circuit types, c is negligible to relevant applications, and in fact scales inversely proportionally to the number of qubits in a single copy, N , for fixed depth circuits.

The purified state may be accessed by,

$$\rho_{\text{pur}}(k) = \frac{\rho^k}{\text{Tr}[\rho^k]}, \quad (1.18)$$

where the denominator acts as a normalisation factor, and for $k = 2$ is the usual definition for the purity of a mixed state. To elucidate the claim that this is the form of the purified state, we expand the spectrum of ρ , assuming that the dominant eigenvector is the ideal state,

$$\rho = \lambda_{\text{id}} |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i |\psi_i\rangle\langle\psi_i|, \quad (1.19)$$

with $\lambda_{\text{id}} > \lambda_i$ for all i , and from normalisation, $\lambda_{\text{id}} + \sum_{i=1}^{2^N-1} \lambda_i = 1$. Inserting this expansion into equation 1.18, we find,

$$\rho_{\text{pur}}(k) = \frac{1}{\mathcal{N}} \left(\lambda_{\text{id}}^k |\psi_{\text{id}}\rangle\langle\psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i^k |\psi_i\rangle\langle\psi_i| \right), \quad (1.20)$$

with normalisation factor $\mathcal{N} = \lambda_{\text{id}}^k + \sum_{i=1}^{2^N-1} \lambda_i^k$. The exponentiation acts to relatively attenuate all error eigenvalues compared to λ_{id} , with the limit in k recovering the pure ideal state, $\lim_{k \rightarrow \infty} \rho_{\text{pur}}(k) = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$.

The remaining question is how can we experimentally access $\rho_{\text{pur}}(k)$? In references [7, 8], it is shown that the expectation value of the purified state with respect to the applied operator $\hat{O}$ is,

$$\langle \hat{O} \rangle_{\text{pur}} = \text{Tr}[\hat{O} \rho_{\text{pur}}(k)] = \frac{\text{Tr}[\hat{O} \rho^k]}{\text{Tr}[\rho^k]}, \quad (1.21)$$

and crucially may be estimated without ever physically preparing $\rho_{\text{pur}}(k)$. Instead the numerator and denominator may be found through preparing k -copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity² providing the numerator,

$$\text{Tr}[\hat{O} \rho^k] = \text{Tr}[\hat{O}^{(i)} \hat{S}^{(k)} \rho^{\otimes k}], \quad (1.22)$$

where $\hat{S}^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^{(i)}$ is the operator acting on the i th copy. The denominator of the corrected expectation value is found by replacing the operator with the identity operator. There is some freedom in the form of the derangement, here we shall give it explicitly as the cyclic permutation of k objects,

$$\hat{S}^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (1.23)$$

Practically, if we can perform this multi-copy derangement, $\hat{S}^{(k)}$, and single-copy operator $\hat{O}^{(i)}$, on k -copies of the noisy state, we can postprocess a collection of circuit repetitions to recover the k th purified expectation value $\langle \hat{O} \rangle_{\text{pur}}$. In the following section we provide two possible approaches for implementing these unitaries and measurements, and give explicit circuit decompositions for both.

Add sample efficiency point

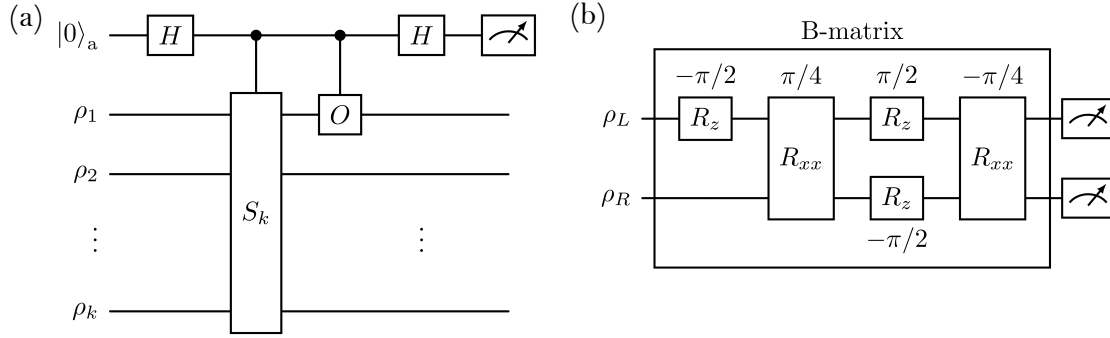


Figure 1.5: Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, $\hat{S}^{(k)}$, over k copies, and controlled operator, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled operator measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle \hat{O} \rangle_{\text{pur}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(\hat{Z}\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle \hat{Z} \rangle_{\text{pur}}$.

Circuits

We discuss two methods of implementing virtual distillation. The first being a general approach able to act on arbitrary Hilbert spaces using Hadamard tests, and the second being more resource efficient, however not simply generalisable to arbitrary sizes of each copy, and to multi-qubit observables.

We begin with the Hadamard test, to build further intuition on virtual distillation based on similar circuit topology discussed in § 1.1.1 for CV-tomography. The circuit for virtual distillation via an ancilla is shown in figure 1.5 (a). The derangement and desired operator, $\hat{O}$, are applied conditioned on the ancilla qubit, with a final measurement only made on the ancilla. The circuit is repeated for N_{shot} times and the outcomes are averaged to recover the estimated real part of the numerator in equation 1.21. The circuit is again repeated N_{shot} times with the controlled operator replaced with the identity operator. Subsequent averaging recovers the denominator, which may then be used to recover $\text{Re}[\langle \hat{O} \rangle_{\text{pur}}]$.

²**Is this footnote on wrong page?** Huggins [8] provides a proof of the identity, which we give an example for the case $k = 2$ in appendix A.9.

The controlled derangement operation may be decomposed into a series of controlled-swap gates [7], which for DV systems may be decomposed to native single and two-qubit gates accessible on the ion trap platform. For CV-systems, the derangement can be decomposed to a series of Fredkin gates [**fredkin_conservative_1982**, **gao_entanglement_2019**], which in turn may be accessible by either controlled-beam-splitters [21, 50] and/or controlled-parity operations [51].

The second method, known as the “ B -matrix” [8], is slightly more resource efficient, not requiring ancilla qubits, however does not simply scale for $k > 2$, and for non- Z measurements. The approach recognises that for $k = 2$ the symmetrised computational observable $\hat{Z}_2 = \frac{1}{2}(\hat{Z}\hat{I} + \hat{I}\hat{Z})$ commutes with the SWAP operator, $\hat{S}^{(2)}$, and consequently so too do both observables required for virtual distillation,

$$[\hat{S}^{(2)}, \hat{Z}_2 \hat{S}^{(2)}] = 0. \quad (1.24)$$

Thus the two observables possess a common eigenbasis and may be simultaneously diagonalised by a single matrix³,

$$\hat{B} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}. \quad (1.25)$$

In the computational eigenbasis, the observable is now given by,

$$\langle \hat{A} \rangle = \text{Tr}[\hat{B}^\dagger \hat{D} \hat{B} \rho], \quad (1.26)$$

$$= \sum_{x \in \{0,1\}^2} d_x p(x), \quad (1.27)$$

where $\hat{A}$ is either $\hat{Z}_2$ or $\hat{S}^{(2)} \hat{Z}_2$, and the second line follows by the definition of the diagonalised operator $\hat{D} = \sum_x d_x |x\rangle \langle x|$, and $p(x) = \langle x | \hat{B} \rho \hat{B}^\dagger | x \rangle$ is the probability of observing computational state $|x\rangle$ after the basis change given by $\hat{B}$.

³Note, we use a different convention than reference [8]. This is done for a simpler decomposition to native gates on our apparatus.

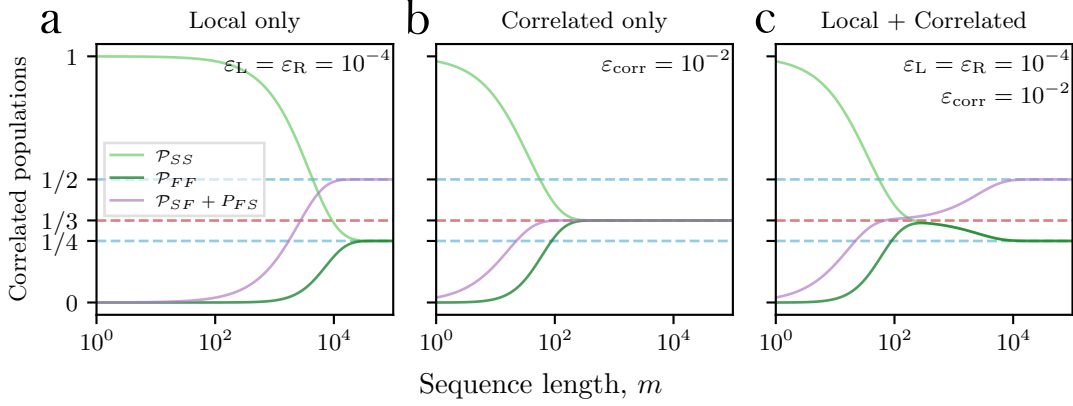


Figure 1.6: Figure: pi-4 MS gate calibration. Figure: Heuristic signal of B-matrix working. Tomography would be best!

As a result, both numerator and denominator from equation 1.21 may be simultaneously measured by computational basis measurements after applying a two-qubit basis rotation⁴. The values are related to computational expectation values by,

$$\text{Tr}[\hat{Z}\rho^2] = \frac{1}{2}(1 - \langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle + \langle \hat{Z} \otimes \hat{Z} \rangle), \quad (1.28)$$

$$\text{Tr}[\rho^2] = \frac{1}{2}(\langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle). \quad (1.29)$$

Figure 1.5 (b) shows the B -matrix decomposition into native gates to our platform. A demonstration of this simple two-copy example of virtual distillation is given in the following section.

1.2.3 Experimental Demonstration

This section begins with experimental details on implementing virtual distillation via the B -matrix decomposed to native R_{xx} gates and single qubit phase shifts. With this protocol we can then study the efficacy of virtual distillation in two different noise regimes — independent depolarisation, and correlated noise due to common random-unitary noise (see § ?? for the noise model). This work provides evidence of the performance of virtual distillation under realistic structured errors originating from fluctuations of a common external field.

⁴Although not discussed here, this method scales simply to Z measurements on copies containing multiple qubits through parallel pairwise rotations [8].

Virtual Distillation on Two Copies

- Minimal demo is two single qubits.
- B circuit demo, consisting of $\pi/4$ gates.
- For single qubit phase push, we use single addressing, push phase on AOD.
- $\pi/4$ gates are walsh 1, calibrated through applying two consecutive and applying same protocol as described in characterisation.
- Give exact numbers for $\pi/4$ gates: loops, power, detuning, duration, fidelity, ramp duration.
- B circuit phases were checked using heuristic method, applying differential single qubit rotation to each.
- Mention here that tomography would have been better?

Virtual Distillation under Correlated Errors

- We target demo of suppression, of controlled injected errors, that are quantified by k-copy RB.
- Give explicit form for suppression of correlated and uncorrelated channels. Look at IRO presentation
- Mention this is found by B circuit applied in Pauli basis and using eqn kcopy pops
- Give motivation of why k-copy RB is particularly relevant to untwirled VD
- Present data in form of single qubit decay rates.
- Give analysis and fits.

1.2.4 Outlook

- Summary of what we have shown experimentally so far.
- Linker sentence to next experimental steps.

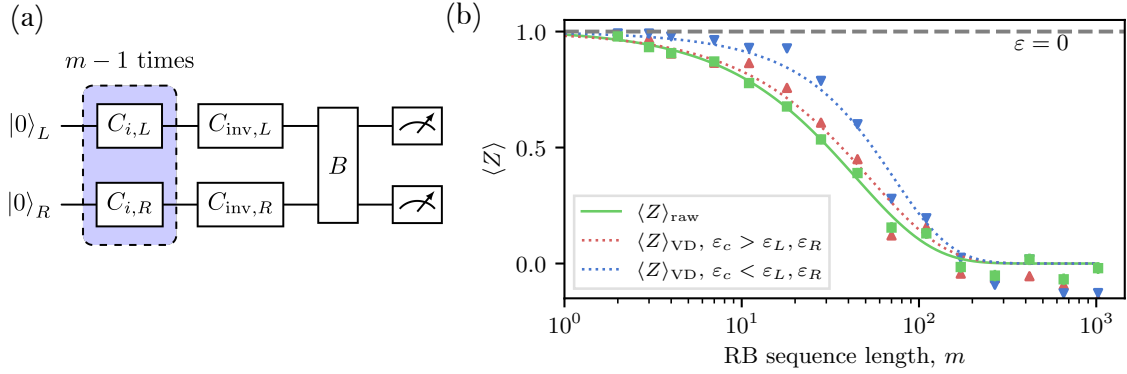


Figure 1.7: Figure: Circuit diagram of RB + VD, exp data for noise floor for two copy B-matrix, with and without spatial correlations

Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Structured errors were injected through a fixed frequency detuning of the 729-nm laser, and the application of a sequence of random Clifford gates. In the case that identical Clifford gates, $C_{i,L} = C_{i,R}$, are applied on both qubits, the detuning error is twirled to a *common noise* channel, while in the case that Clifford gates are sampled independently for both copies, the detuning is twirled to local depolarisation channels. The local and correlated injected error-per-Clifford for both regimes are quantified using k -copy RB. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective reduction in error-per-Clifford by applying the virtual distillation circuit. The green points correspond to the raw single-qubit decay due to the physical errors, the solid green line is a fit of this decay to measure a total error-per-Clifford. The red and blue points correspond to the error mitigated Z observables for the dominant correlated errors and local depolarising errors respectively. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation ??, and the error-per-Cliffords measured by k -copy RB.

Extending to > 2 copies

- plan for controlled swap implementation, requirements for this.
 - How many gates, what fidelity we need to ensure tot-err < 1 for derangement side.
 - Given 98% how many copies could we go to? Can we estimate shot numbers here?
- Summary in a table would be nice.
- copy number | derangement gates | circuit fidelity | expected sampling overhead

Virtual Distillation on Hybrid Systems

- Why important to develop, applications in simulations of QO demo, Feynman, gauge theories.

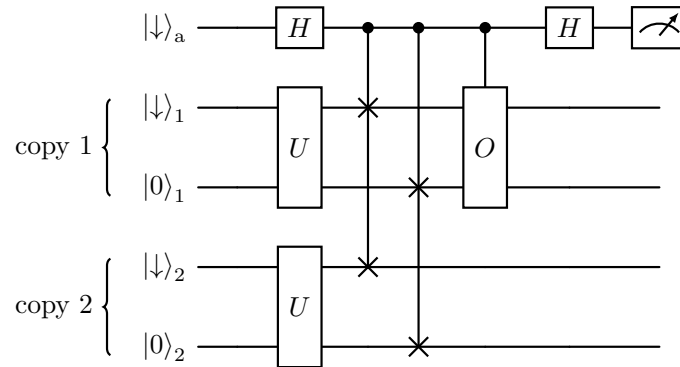


Figure 1.8: Figure: Fredkin gate on oscillator, and how to compose derangement transversally. Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates.

- Cite Quantum virtual cooling
- How it looks on a thermal state and on various quantum states (sqz, coherent, fock)
- How we can implement c-swap using CV gate set.

Limitations

- Never actually remove entropy from system, so we are limited to total error rate < 1 , unlike QEC. This places us firmly in NISQ applications.
- Coherent and structured errors. Mitigation is twirling.
- Coherent noise floor
- Slow readout if mapping to single ancilla?

1.3 Conclusions

- Summary of chapter, saying this gives our future directions.
- Future planned hardware extensions will give more options for CV.

Standing wave extension

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states [52, 53]. We shall see that the language used for this analysis of single qubit

gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These

two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.6 Projectors of $Cl_1^{\otimes 2}$

We outline the method of recovering the α parameters described in section ??? using projectors onto the irrep spaces. A special thanks to Iver Øvergaard who performed this analysis applied to the *common noise* channel and gave permission for reproducing the results here **IRO please check you are happy with me to include this appendix.**

The decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes R)$, where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and

$R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (\text{A.10})$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table ?? . However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (\text{A.11})$$

$$\Pi_{dt} = D - T, \quad (\text{A.12})$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (\text{A.13})$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (\text{A.14})$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (\text{A.15})$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (\text{A.16})$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (\text{A.17})$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (\text{A.18})$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (\text{A.19})$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Substituting these traces into equation A.10

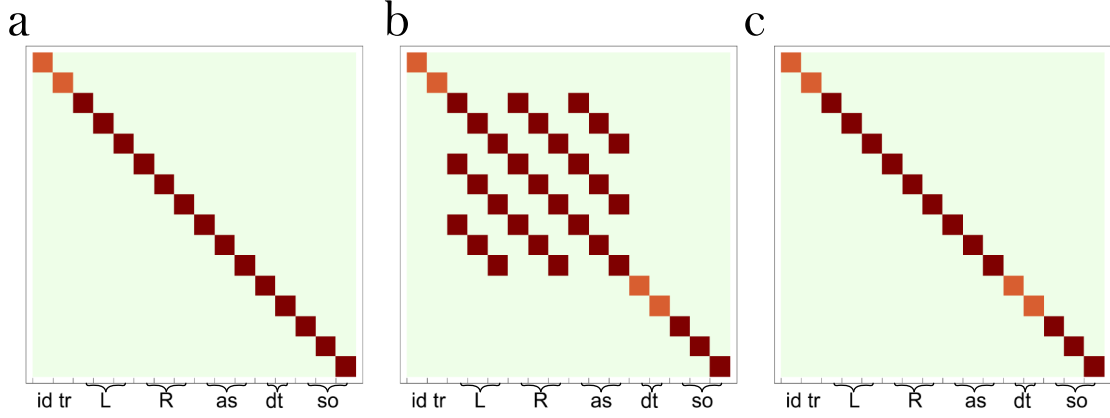


Figure A.1: Liouville operator representation of error channels after twirling over $Cl_1^{\otimes 2}$, ordered by the irrep basis (Schur basis). (a) *Common noise*, (b) *Continuous ZZ*, (c) *Stochastic ZZ*. Orange blocks represent an entry of 1, green represent 0, and red represent a decay with dependence on m .

and using equation A.11 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (\text{A.20})$$

$$\alpha_{tr} = 1, \quad (\text{A.21})$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (\text{A.22})$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa). \quad (\text{A.23})$$

A.7 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.24})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table ???. We can show the qualitative difference

in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the *ZZ*-type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure ?? plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled *ZZ*-type noise, given by equation A.24. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant *ZZ*-type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic *ZZ* noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.25})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic *ZZ* error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.26})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and *ZZ*-type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure

of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section ?? . Analogous protocols to this is described by character RB [54], and tomography via RB [55].

A.8 Magnus expansion

The unitary evolution due to a time dependent Hamiltonian may be written in the ordered, short-time limit as,

$$U(t) = \lim_{N \rightarrow \infty} e^{\frac{-i}{\hbar} \hat{H}(t_N) \Delta t} \dots e^{\frac{-i}{\hbar} \hat{H}(t_2) \Delta t} e^{\frac{-i}{\hbar} \hat{H}(t_1) \Delta t} = e^{A(t)}, \quad (\text{A.27})$$

where $\Delta t = t/N$, and $t_n = n\Delta t$, and $A(t)$ is a time dependent exponent, which is expanded in the Magnus construction to [34],

$$A(t) = \sum_{k=1}^{\infty} A_k(t). \quad (\text{A.28})$$

If the Hamiltonian at different times does not commute, then the first three $A_k(t)$ terms are given by,

$$A_1(t) = \frac{-i}{\hbar} \int_0^t dt_1 \hat{H}(t_1) \quad (\text{A.29})$$

$$A_2(t) = -\frac{1}{2\hbar^2} \int_0^t \int_0^{t_1} dt_1 dt_2 [\hat{H}(t_1), \hat{H}(t_2)] \quad (\text{A.30})$$

$$A_3(t) = \frac{i}{6\hbar^3} \int_0^t \int_0^{t_1} \int_0^{t_2} dt_1 dt_2 dt_3 \left([\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]] + [\hat{H}(t_3), [\hat{H}(t_2), \hat{H}(t_1)]] \right). \quad (\text{A.31})$$

This may be understood as the continuous analogy of the Baker-Campbell-Hausdorff formula. It is particularly useful in this thesis to understand the origin of the geometric phase in the MS gate scheme, and the non-linear bosonic interactions generated in the “two-SDF” protocol given in § 1.1.1.

A.8.1 “Two-SDF” dynamics

Here we analyse the “two-SDF” Hamiltonian given in equation A.32 using the Magnus expansion. We recover effective Hamiltonians for both Gaussian interactions, as well as for tri-squeezing.

The “two-SDF” Hamiltonian is reproduced for convenience here,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (\text{A.32})$$

$$= \frac{\hbar\Omega_\alpha}{2} \hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (\text{A.33})$$

$$+ \frac{\hbar\Omega_\beta}{2} \hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right). \quad (\text{A.34})$$

The first three terms of the corresponding Magnus expansion are found by inserting $H_{2\text{SDF}}$ into equations A.29. We shall later rationalise that with certain values of SDF magnitudes, detunings, and phases, we may arbitrarily suppress the higher order terms.

The A_1 term appears as separable fast oscillations in phase space due to each SDF,

$$A_1(t) = -i \sum_{i \in \{\alpha, \beta\}} \left(\frac{\Omega_i \hat{\sigma}_i}{\Delta_i} \sin\left(\frac{\Delta_i t}{2}\right) \left(\hat{a}_i e^{i\left(\frac{\Delta_i t}{2} + \phi_i\right)} + \hat{a}_i^\dagger e^{-i\left(\frac{\Delta_i t}{2} + \phi_i\right)} \right) \right). \quad (\text{A.35})$$

This is effectively the first-order only dynamics of the displacement operator leading to excursions in phase-space. The effects of this term may be entirely suppressed at certain gate durations, $A_1(T) = 0$, where T is an integer multiple of both $\tau_\alpha = 2\pi/\Delta_\alpha$ and $\tau_\beta = 2\pi/\Delta_\beta$. Excursions may also be made arbitrarily small by ensuring $\Omega_i \ll \Delta_i$.

The commutator in A_2 term may be expanded to reveal four relevant components,

$$A_2(t) = S_\alpha(t) + S_\beta(t) + X_{\text{pair}}(t) + X_{\text{exch}}(t), \quad (\text{A.36})$$

where S denotes “self-terms”, and X “cross-terms” of the two simultaneously applied SDFs. The self-terms, with the form,

$$S_i(t) = \frac{i\Omega_i^2}{4} \left(\frac{\sin(\Delta_i t)}{\Delta_i^2} - \frac{t}{\Delta_i} \right) \quad (\text{A.37})$$

are recognisable from section ?? as geometric phase factors. Taking the assumption that $t \gg \frac{1}{\Delta_i}$, we get a linearly growing geometric phase term, $S_i(t) \approx \frac{-i\Omega_i^2 t}{4\Delta_i}$. In the

case that this geometric phase acts globally on the system, they may be ignored. If the accumulated phase is spin, or oscillator state dependent, then interesting dynamics may be observed, however these effects are out of the scope of this thesis. The remaining cross-terms contain the 2nd order non-linear bosonic interactions we are interested in emulating. To recover the general family of non-linear bosonic interactions given in table 1.1, motional phases are set to $\phi_\alpha = 0$, $\phi_\beta = \phi$, and spin-conditioning is assumed to be the anti-commuting pair: $\sigma_\alpha = \sigma_x$, $\sigma_\beta = \sigma_y$. We set the detunings of the two SDF to be $\Delta_\beta = m\Delta_\alpha$, where m is some integer. Expanding the commutator of the cross terms we find,

$$X_{\text{pair}} =, \quad (\text{A.38})$$

$$X_{\text{exch}} =, \quad (\text{A.39})$$

$$(\text{A.40})$$

where... We see now the freedom to “tune” the desired interaction, if we set $m = +1$, we resonantly drive..., alternatively if we set $m = -1$, we recover... We can convert from the generator form, $A(T)$, to familiar effective time-independent Hamiltonian, H_{eff} for a given time, T , by $\hat{H}_{\text{eff}} = \frac{i\hbar}{T}A(T)$.

For $m = +1$, we find that the effective Hamiltonian is,

$$H_{\text{eff,exch}} = \dots \quad (\text{A.41})$$

which for a single mode, $a^\dagger = b^\dagger$, is the parity operation, and for two modes, is the beam splitter.

For $m = -1$, we find,

$$H_{\text{eff,pair}} = \dots \quad (\text{A.42})$$

which for a single mode, $a^\dagger = b^\dagger$, is the squeezing operation, and for two modes, is the two-mode squeezing.

Similar arguments hold for $m = 2$, where now the A_2 cross terms appear off-resonant by..., and the on resonant cross terms in A_3 coherently sum over time yielding effective 3rd order bosonic operators, such as the tri-squeezing interaction.

Higher order error term scaling...

A.9 Derangement for $k = 2$

Huggins [8] provides a proof of the identity given in equation 1.23, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.43})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (\text{A.44})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (\text{A.45})$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.46})$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \quad (\text{A.47})$$

$$= \text{Tr}[\hat{O} \rho^2], \quad (\text{A.48})$$

with the last equality following from the definition of matrix multiplication.

References

- [1] O. Băzăvan et al. *Squeezing, trisqueezing, and quadsqueezing in a spin-oscillator system*. en. arXiv:2403.05471 [physics, physics:quant-ph]. 2024.
- [2] S. Saner et al. *Generating arbitrary superpositions of nonclassical quantum harmonic oscillator states*. arXiv:2409.03482 [quant-ph]. 2024.
- [3] Oana Bazavan. “Easy as XYZ: Exploring non-commuting coherent interactions in trapped ions”. en. In: ().
- [4] Sebastian Saner. *Quantum Control of Hybrid Spin-Oscillator Systems*. en. Springer Theses. Cham: Springer Nature Switzerland, 2026.
- [5] Kai Shinbrough et al. *Applications of Spin-Dependent Generalized Squeezing in Hybrid Spin-Oscillator Quantum Processors*. arXiv:2608.04197 [quant-ph]. 2026.
- [6] Jordan Cotler et al. “Quantum Virtual Cooling”. In: *Phys. Rev. X* 9.3 (2019). arXiv:1812.02175 [quant-ph].
- [7] Bálint Koczor. “Exponential Error Suppression for Near-Term Quantum Devices”. In: *Phys. Rev. X* 11.3 (2021).
- [8] William J. Huggins et al. “Virtual Distillation for Quantum Error Mitigation”. In: *Phys. Rev. X* 11.4 (2021).
- [9] Seth Lloyd and Samuel L. Braunstein. “Quantum Computation over Continuous Variables”. In: *Phys. Rev. Lett.* 82.8 (1999).
- [10] Yuan Liu et al. “Hybrid Oscillator-Qubit Quantum Processors: Instruction Set Architectures, Abstract Machine Models, and Applications”. In: *PRX Quantum* 7.1 (2026).
- [11] Lukas Brenner et al. *Factoring an integer with three oscillators and a qubit*. en. arXiv:2412.13164 [quant-ph]. 2024.
- [12] Timo Hillmann et al. “Universal Gate Set for Continuous-Variable Quantum Computation with Microwave Circuits”. In: *Phys. Rev. Lett.* 125.16 (2020).
- [13] Alexandre Blais et al. “Circuit quantum electrodynamics”. In: *Rev. Mod. Phys.* 93.2 (2021).
- [14] J. Majer et al. “Coupling superconducting qubits via a cavity bus”. en. In: *Nature* 449.7161 (2007).
- [15] Xiaozhou Pan, Pengtao Song, and Yvonne Y. Gao. “Continuous-Variable Quantum Computation in Circuit QED”. en. In: *Chinese Phys. Lett.* 40.11 (2023).
- [16] Alec Eickbusch et al. “Fast universal control of an oscillator with weak dispersive coupling to a qubit”. en. In: *Nat. Phys.* 18.12 (2022).
- [17] A. M. Kaufman, B. J. Lester, and C. A. Regal. “Cooling a Single Atom in an Optical Tweezer to Its Quantum Ground State”. In: *Phys. Rev. X* 2.4 (2012).

- [18] Vincent Lienhard et al. “Generation of Motional Squeezed States for Neutral Atoms in Optical Tweezers”. In: *Phys. Rev. Lett.* 135.25 (2025).
- [19] Jaeyong Hwang et al. “Hybrid qubit-oscillator module from motional states of two interacting atoms”. In: *Phys. Rev. Lett.* 137.11 (2026). arXiv:2512.06429 [quant-ph].
- [20] R. T. Sutherland and R. Srinivas. “Universal hybrid quantum computing in trapped ions”. en. In: *Phys. Rev. A* 104.3 (2021). arXiv:2105.05768 [physics, physics:quant-ph].
- [21] H. C. J. Gan et al. “Hybrid Quantum Computing with Conditional Beam Splitter Gate in Trapped Ion System”. en. In: *Phys. Rev. Lett.* 124.17 (2020).
- [22] C. Flühmann et al. “Encoding a qubit in a trapped-ion mechanical oscillator”. en. In: *Nature* 566.7745 (2019).
- [23] D. J. Heinzen and D. J. Wineland. “Quantum-limited cooling and detection of radio-frequency oscillations by laser-cooled ions”. In: *Phys. Rev. A* 42.5 (1990).
- [24] S. C. Burd et al. “Quantum amplification of mechanical oscillator motion”. en. In: *Science* 364.6446 (2019).
- [25] S. C. Burd et al. “Quantum amplification of boson-mediated interactions”. en. In: *Nat. Phys.* 17.8 (2021).
- [26] Martín Drechsler et al. “State-dependent motional squeezing of a trapped ion: Proposed method and applications”. In: *Phys. Rev. A* 101.5 (2020).
- [27] Philip Leindecker et al. *State-dependent Gaussian gate set using an optical tweezer for trapped ions*. arXiv:2606.31864 [quant-ph]. 2026.
- [28] Anders Sorensen and Klaus Molmer. “Entanglement and quantum computation with ions in thermal motion”. en. In: *Phys. Rev. A* 62.2 (2000). arXiv:quant-ph/0002024.
- [29] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. en. ISBN: 9780511976667. 2010.
- [30] Scott Aaronson and Daniel Gottesman. “Improved simulation of stabilizer circuits”. In: *Phys. Rev. A* 70.5 (2004).
- [31] Samuel L. Braunstein and Peter van Loock. “Quantum information with continuous variables”. In: *Rev. Mod. Phys.* 77.2 (2005).
- [32] Axel M. Eriksson et al. “Universal control of a bosonic mode via drive-activated native cubic interactions”. en. In: *Nat Commun* 15.1 (2024).
- [33] S. Saner et al. *Real-Time Observation of Aharonov-Bohm Interference in a $\mathbb{Z}_2$ Lattice Gauge Theory on a Hybrid Qubit-Oscillator Quantum Computer*. en. arXiv:2507.19588 [quant-ph]. 2025.
- [34] Wilhelm Magnus. “On the exponential solution of differential equations for a linear operator”. en. In: *Communications on Pure and Applied Mathematics* 7.4 (1954).
- [35] Christa Flühmann and Jonathan P. Home. “Direct characteristic-function tomography of quantum states of the trapped-ion motional oscillator”. en. In: *Phys. Rev. Lett.* 125.4 (2020). arXiv:1907.06478 [physics, physics:quant-ph].
- [36] R. Cleve et al. “Quantum algorithms revisited”. In: *Proc. R. Soc. A* 454.1969 (1998).

- [37] E. Wigner. “On the Quantum Correction For Thermodynamic Equilibrium”. In: *Phys. Rev.* 40.5 (1932).
- [38] K. E. Cahill and R. J. Glauber. “Density Operators and Quasiprobability Distributions”. In: *Phys. Rev.* 177.5 (1969).
- [39] K. Vogel and H. Risken. “Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase”. In: *Phys. Rev. A* 40.5 (1989).
- [40] Hsiang-Yu Lo et al. “Spin–motion entanglement and state diagnosis with squeezed oscillator wavepackets”. en. In: *Nature* 521.7552 (2015). Number: 7552.
- [41] Kevin He et al. “Efficient multimode Wigner tomography”. en. In: *Nat Commun* 15.1 (2024).
- [42] Christophe H. Valahu et al. “Benchmarking Bosonic Modes for Quantum Information with Randomized Displacements”. In: *PRX Quantum* 5.4 (2024).
- [43] A. Walther et al. “Controlling Fast Transport of Cold Trapped Ions”. In: *Phys. Rev. Lett.* 109.8 (2012).
- [44] L. Luo et al. “Protocols and techniques for a scalable atom–photon quantum network”. en. In: *Fortschritte der Physik* 57.11-12 (2009).
- [45] M. Valentini et al. “Demonstration of Two-Dimensional Connectivity for a Scalable Error-Corrected Ion-Trap Quantum Processor Architecture”. In: *Phys. Rev. X* 15.4 (2025).
- [46] V. M. Schäfer et al. “Fast quantum logic gates with trapped-ion qubits”. en. In: *Nature* 555.7694 (2018). Number: 7694.
- [47] Zhenyu Cai et al. “Quantum error mitigation”. In: *Rev. Mod. Phys.* 95.4 (2023).
- [48] Bálint Koczor. “The Dominant Eigenvector of a Noisy Quantum State”. In: *New J. Phys.* 23.12 (2021). arXiv:2104.00608 [quant-ph].
- [49] Jonathan Foldager and Bálint Koczor. “Can shallow quantum circuits scramble local noise into global white noise?” en. In: *J. Phys. A: Math. Theor.* 57.1 (2024).
- [50] Honggi Jeon et al. “Two-Mode Bosonic State Tomography with Single-Shot Joint-Parity Measurement of a Trapped Ion”. en. In: *PRX Quantum* 6.4 (2025).
- [51] Wonhyeong Choi et al. *Multimode Fock-State Measurements using Dispersive Shifts in a Trapped Ion*. en. arXiv:2601.04591 [quant-ph]. 2026.
- [52] M. D. Bowdrey et al. “Fidelity of Single Qubit Maps”. en. In: *Physics Letters A* 294.5-6 (2002). arXiv:quant-ph/0201106.
- [53] Michael A Nielsen. “A simple formula for the average gate fidelity of a quantum dynamical operation”. In: *Physics Letters A* 303.4 (2002).
- [54] Jonas Helsen et al. “A new class of efficient randomized benchmarking protocols”. en. In: *npj Quantum Inf* 5.1 (2019).
- [55] Shelby Kimmel et al. “Robust Extraction of Tomographic Information via Randomized Benchmarking”. en. In: *Phys. Rev. X* 4.1 (2014).